# COMPARATIVE STUDY OF POPULAR NO-CODE AND LOW-CODE APPLICATION DEVELOPMENT PLATFORMS IN ZOHO

## Aim

To conduct a comparative study of no-code and low-code application development using the Zoho platform, with special focus on Zoho Creator, and to analyze its features, development methods, ease of use, customization, automation, integration, scalability, advantages, and limitations.

## Objectives

- To understand the concepts of no-code and low-code application development.
- To study the role of Zoho Creator in application development.
- To understand the difference between no-code and low-code development within Zoho Creator.
- To study the drag-and-drop application development approach.
- To analyze form, report, page, and workflow creation.
- To study database and data management capabilities.
- To analyze integration with other Zoho applications and third-party services.
- To understand workflow automation using Zoho Creator.
- To study the use of Deluge scripting for advanced customization.
- To analyze the advantages and limitations of Zoho Creator.
- To identify suitable applications that can be developed using the Zoho platform.

## Problem Statement

Traditional application development requires programming knowledge, database management, server configuration, testing, and deployment. This can increase development time and require skilled developers.

Small businesses and organizations may need applications for inventory management, employee management, customer management, project tracking, data collection, approvals, and reporting, but they may not have enough programming resources.

Zoho Creator addresses this problem by providing a visual development environment where users can create applications using drag-and-drop components, forms, reports, pages, workflows, and integrations. For advanced requirements, developers can use Deluge scripting, APIs, serverless functions, and other developer tools.

## Zoho Creator as a No-Code and Low-Code Platform

Zoho Creator can be considered a low-code application development platform with strong no-code capabilities. Users can create many applications without writing traditional programming code, while developers can add scripts and custom logic when necessary.

## No-Code Approach

- Drag-and-drop form builder
- Visual page builder
- Pre-built templates
- Visual workflows

- Reports and dashboards
- Ready-made integrations
- Application components
- Automated processes

This approach is suitable for users with limited programming knowledge.

## Low-Code Approach

- Deluge scripting
- APIs
- Serverless functions
- Custom components
- Custom integrations
- Advanced workflows
- Custom business logic

This provides greater flexibility than a purely no-code platform.

## Major Features of Zoho Creator

## a. Drag-and-Drop Application Builder

Zoho Creator provides visual development tools that allow users to create application components without manually writing large amounts of code. Forms, pages, workflows, and reports can be designed using visual interfaces.

## b. Database and Forms

Users can create forms to collect and manage business data and define application data structures.

## c. Workflow Automation

Business processes can be automated using workflows and Blueprints for notifications, approvals, updates, and scheduled operations.

## d. Reports and Dashboards

Zoho Creator provides reporting and dashboard capabilities for analyzing application data and can integrate with Zoho Analytics.

## e. Integration

Zoho Creator supports integration with Zoho applications and external services through integrations, connectors, and APIs.

## f. Mobile Application Development

Applications created in Creator can be made available across web and mobile environments, including Android and iOS.

## g. AI Assistance

Zoho Creator includes AI-assisted features that can help with application creation, form generation, script generation, and development tasks.

## Comparative Study: No-Code vs Low-Code in Zoho Creator

| Parameter | No-Code Approach | Low-Code Approach |
|---|---|---|
| Programming | Little or no coding | Limited coding when required |
| Development Method | Drag-and-drop | Visual development + scripting |
| User Type | Business users / beginners | Developers and advanced users |
| Form Creation | Visual | Visual + custom logic |
| Workflow | Visual workflows | Visual workflows + Deluge |
| Customization | Moderate | High |
| Database | Built-in | Built-in + advanced integration |
| API Integration | Connectors | APIs + custom integrations |

## Application Development Process in Zoho Creator

**Identify Requirement**

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**Design Application**

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**Create Forms and Database**

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**Create Pages and Reports**

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**Configure Workflows**

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**Add Integrations**

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**Add Deluge Code if Required**

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**Test Application**

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**Deploy Application**

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**Monitor and Maintain**

## Example Application – Inventory Management System

A simple Inventory Management System can be developed using Zoho Creator. The application can contain Product, Supplier, Inventory Report, and Workflow modules.

- Product Form: Product ID, Product Name, Category, Quantity, Price, and Supplier.

- Supplier Form: Supplier ID, Supplier Name, Contact, and Address.
- Inventory Report: Available products, low-stock products, total inventory, and product-wise quantity.
- Workflow: When stock falls below a specified level, generate an alert; when new stock is added, update inventory; when a product is sold, reduce available quantity.

This type of application can initially be created using visual components and later extended using Deluge scripting for advanced business rules.

## Advantages of Zoho Creator

- Reduces traditional programming requirements.
- Provides drag-and-drop development.
- Speeds up application development.
- Supports both no-code and low-code development.
- Provides database and form management.
- Supports workflow automation.
- Provides reports and dashboards.
- Supports Zoho and third-party integrations and APIs.
- Supports web and mobile applications.
- Provides developer tools for advanced customization.
- Supports AI-assisted development.
- Can be used by business users and professional developers.

## Limitations

- Advanced applications may still require programming knowledge.
- Developers need to learn Deluge for advanced customization.
- Some advanced features depend on the selected subscription plan.
- Applications are strongly connected to the Zoho ecosystem.
- Complex requirements may require APIs or custom development.
- Migration to another platform may require additional effort.
- Large applications require proper architecture and database planning.

## Applications of Zoho Creator

- Inventory Management Systems
- Employee Management Systems
- Customer Management Applications
- Project Management Systems
- Attendance Applications
- Sales Management Systems
- Expense Management Applications
- Approval Systems
- Service Management Applications
- School and College Management Systems
- Online Data Collection Applications
- Business Dashboards
- Customer Portals

- Workflow Automation Systems

## Result

The comparative study shows that Zoho Creator provides both no-code and low-code capabilities within a single application development environment. Users with limited programming knowledge can create applications using forms, reports, pages, drag-and-drop components, and visual workflows.

For more complex requirements, developers can use Deluge scripting, APIs, serverless functions, and custom integrations to extend application functionality.

The study also shows that Zoho Creator is particularly useful for business applications, data management systems, workflow automation, dashboards, portals, and mobile applications.